SIMATS ENGINEERING

ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

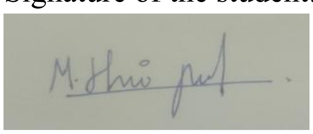
Total Marks: 50

Code of Conduct

I Hariprasath(192421246) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1.Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Definition

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, software, analytics, and artificial intelligence over the internet ("the cloud"). It allows users to access resources on demand without owning or maintaining physical infrastructure. Cloud services are provided on a **pay-as-you-go** basis, reducing costs and increasing flexibility.

Characteristics

1. On-Demand Self-Service

Users can provision computing resources such as virtual machines, storage, and databases whenever required without interacting directly with the service provider. This enables faster deployment of applications.

2. Broad Network Access

Cloud services are available over the internet and can be accessed from various devices such as laptops, desktops, tablets, and smartphones using standard web browsers or applications.

3. Resource Pooling

Cloud providers pool computing resources to serve multiple customers using a multi-tenant model. Resources are dynamically allocated and reassigned according to user demand, improving efficiency.

4. Rapid Elasticity and Scalability

Cloud resources can be quickly increased or decreased depending on workload. Businesses can handle sudden increases in traffic without purchasing additional hardware.

Difference from Traditional On-Premise Computing

Unlike on-premise computing, which requires organizations to purchase, install, and maintain physical hardware, cloud computing provides resources over the internet with lower upfront costs, automatic updates, and greater flexibility.

2.Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Cloud computing evolved through several technological advancements over the years.

1. Mainframe Computing (1950s–1970s)

Large mainframe computers were shared among multiple users through terminals. These systems introduced the concept of sharing computing resources.

2. Personal Computers (1980s)

The introduction of affordable personal computers allowed individuals and businesses to perform computing tasks independently.

3. Networking (1990s)

Local Area Networks (LAN) and Wide Area Networks (WAN) enabled computers to communicate and share files, printers, and applications.

4. Internet Evolution

The rapid growth of the internet allowed users worldwide to access websites, applications, and online services from anywhere.

5. Virtualization

Virtualization technology enabled multiple virtual machines to run on a single physical server. This improved server utilization, reduced costs, and simplified resource management.

6. Web-Based Software

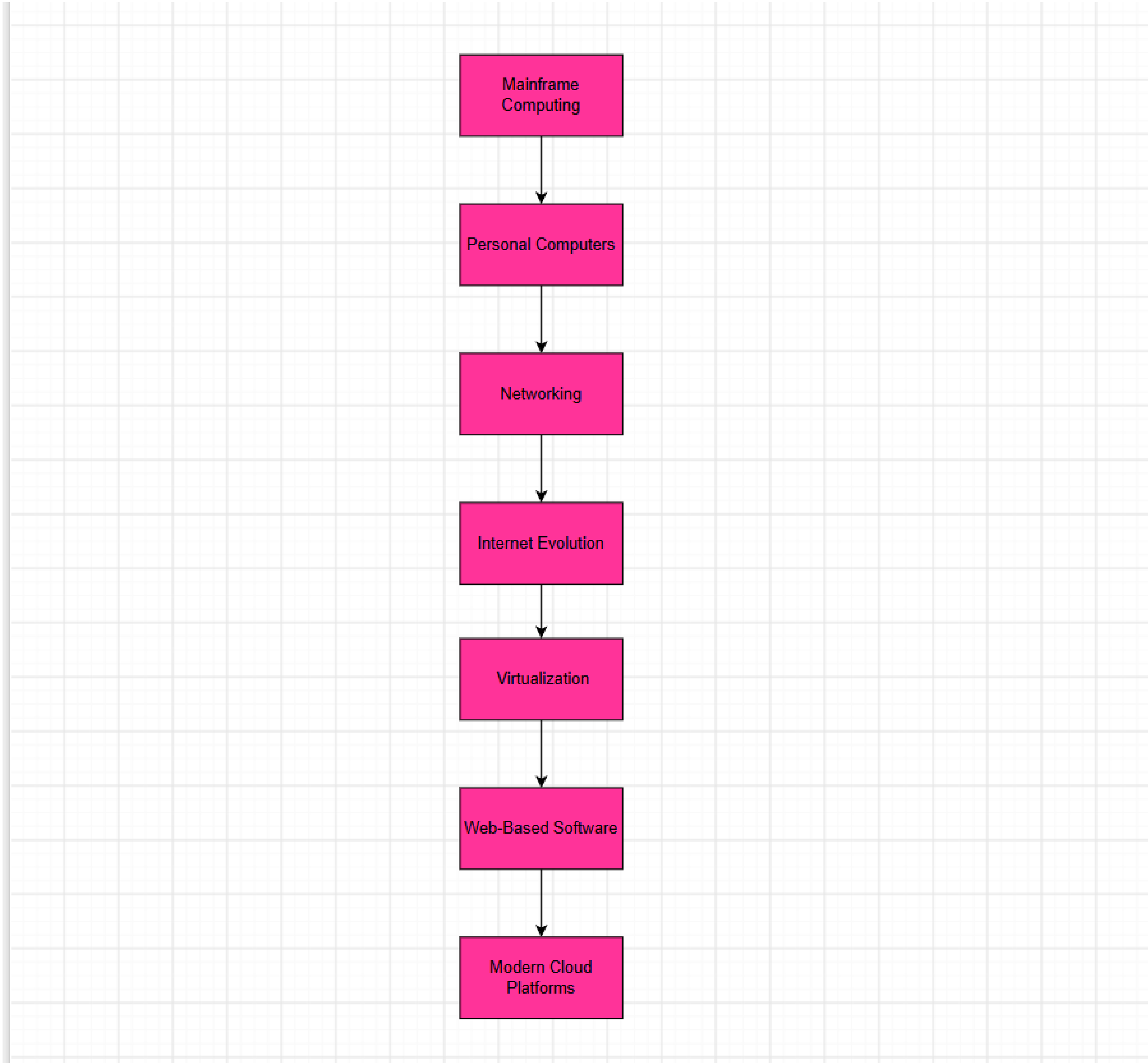
Applications moved from desktop installations to browser-based services such as email, collaboration tools, and online storage.

7. Modern Cloud Platforms

Companies such as **Amazon Web Services (AWS)**, **Microsoft Azure**, and **Google Cloud Platform (GCP)** introduced cloud services that provide computing, storage, databases, networking, and AI resources over the internet.

Conclusion

The evolution from mainframes to internet-based software, combined with virtualization, enabled today's cloud platforms to offer scalable, reliable, and cost-effective services to users worldwide.


Mainframe Computing
Personal Computers
Networking
Internet Evolution
Virtualization
Web-Based Software
Modern Cloud Platforms

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Role of Server Virtualization

Server virtualization is a technology that divides one physical server into multiple virtual servers called **Virtual Machines (VMs)**. Each VM runs its own operating system and applications independently.

Advantages

- Maximizes hardware utilization.
- Reduces infrastructure costs.
- Enables rapid deployment of virtual machines.
- Simplifies backup and disaster recovery.
- Improves scalability and resource management.

Virtualization is one of the core technologies that makes cloud computing possible.

Difference between Virtualization and Cloud Computing

Virtualization	**Cloud Computing**
It is a technology.	It is a service delivery model.
Creates multiple virtual machines on one physical server.	Delivers computing resources over the internet.
Focuses on efficient hardware utilization.	Focuses on providing services to users.
Can exist without cloud computing.	Often uses virtualization to provide services.
Managed within an organization's infrastructure.	Managed by cloud service providers or private cloud administrators.

4.Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications

Feature	**SOAP**	**REST**
Communication Style	Protocol-based	Architectural style
Data Exchange	XML only	JSON, XML, HTML, Plain Text
Performance	Slower due to XML	Faster due to lightweight JSON
Complexity	More complex	Simple and easy to implement
Security	Built-in WS-Security	Uses HTTPS, OAuth, JWT
Flexibility	Less flexible	Highly flexible
Cloud Suitability	Enterprise applications requiring high security	Web, mobile, and cloud applications

5.Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Cloud computing provides different service models depending on the level of control required by users.

Service Model	Description	User Manages	Provider Manages	Example
IaaS (Infrastructure as a Service)	Provides virtual machines, storage, and networking.	Operating system, applications, and data.	Hardware, networking, and virtualization.	Amazon EC2
PaaS (Platform as a Service)	Provides a platform for developing, testing, and deploying applications.	Applications and data.	Infrastructure, operating system, and runtime.	Google App Engine
SaaS (Software as a Service)	Provides ready-to-use software over the internet.	User settings and data.	Everything else.	Microsoft 365
XaaS (Anything as a Service)	Refers to any IT service delivered through the cloud, including storage, networking, databases, and AI services.	Depends on the service.	Entire cloud service infrastructure.	Microsoft Azure